DG-57
Service Manual

SWITCHED POWER SUPPLY 24V ADJUSTMENT AND VERIFICATION

Procedure description
This procedure describes how adjust the electronic values of the switched power supply in order to make sure that the equipment is ready to be used. *Note: Document REGD-0000791_v4 replaces document REGD-0000791_v3*

EQUIPMENT

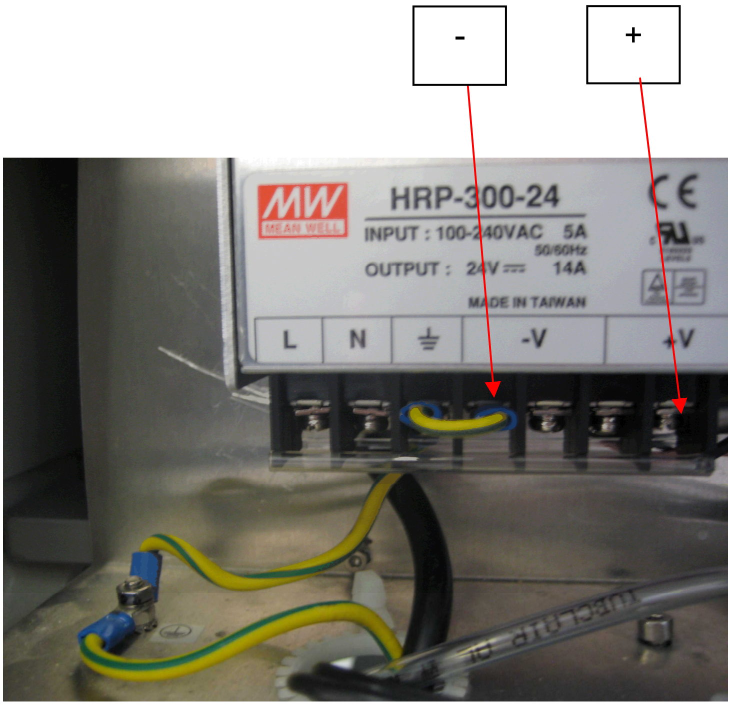
1.	Multimeter.

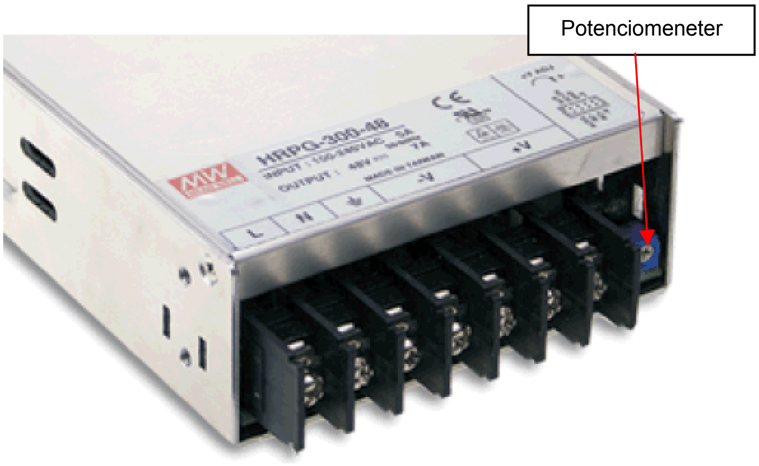
SET UP

1.	- - -

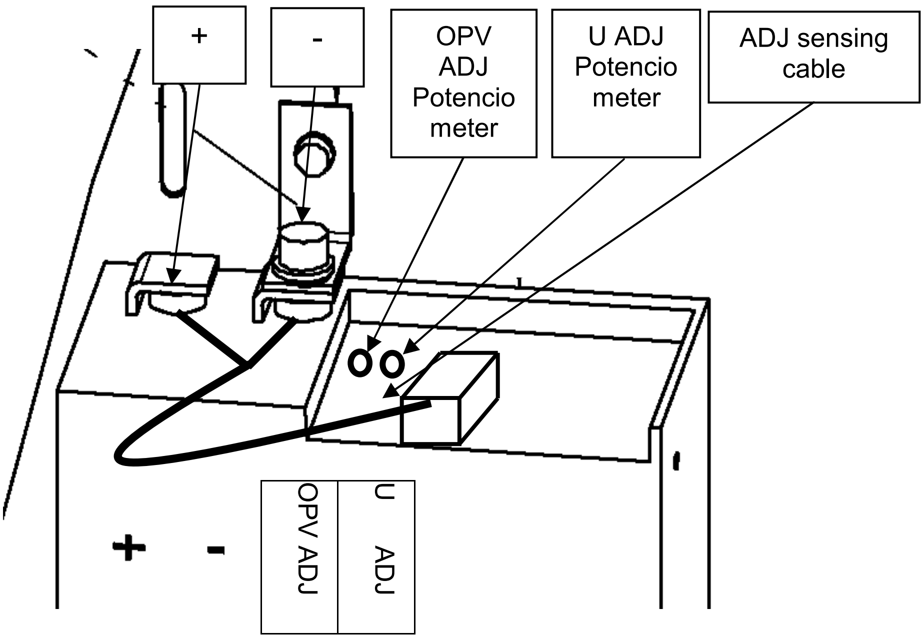
!	**CAUTION!** Short-circuit between any powered part and ground, may damage the instrument

PROCEDURE 1 (current power supply model)

Process	Expected Results
Checking the SPS source 1. Measure the voltage BETWEEN "+" and "-" with the multimeter  Picture 1	- - -
2. Check if the measured value reaches expected result. If required, adjust SPS.	24V± 1%
Adjusting SPS output voltage. 3. Switch off the equipment.	---
4. Disconnect all cables from '+' and '-' SPS output terminals avoiding PCBs to be damaged.	---

Process	Expected Results
5. Switch on the equipment.	---
6. Adjust potentiometer (see Picture 2) to adjust SPS output voltage:  Picture 2	24V± 1%

PROCEDURE 2 (old power supply model)

Process	Expected Results
Checking the SPS source 1. Measure the voltage BETWEEN "+" and "-" with the multimeter  Picture 3	---

Process	Expected Results
2. Check if the measured value reaches expected result. If required, adjust SPS.	23,95 ± 0,05 V
Adjusting the SPS source 3. Switch off the equipment.	---
4. Disconnect all cables from '+' and '-' terminals except for ADJ sensing cable. (see Picture 1).	---
5. Adjust the potentiometer OVP ADJ in a clockwise direction until it can go no further (see Picture 1). OVP means Overvoltage Protection	---
6. Connect the power source to the mains network and check that there is voltage in the output.	---
7. Adjust potentiometer U ADJ to 25,2 ± 0,1 V (see Picture 1).	
8. Adjust the OVP ADJ counter-clockwise, very slowly, until the power supply goes to approximately 0V	---
9. With the power supply cut, rotate U ADJ counter-clockwise, turning several times if possible	
10. Disconnect the network supply. **NOTE:** Upon reaching this point the supply will remain internally inhibited for around five minutes	---
11. Connect the supply. Adjust U ADJ clockwise until it is possible to check that the power supply is cut off within the following margins. If this is not the case, go back to step 6.	25,2 ± 0,1 V
12. With the power supply cut, rotate U ADJ once, counter-clockwise.	---
13. Disconnect the network supply. **NOTE:** Upon reaching this point the supply will remain internally inhibited for around five minutes	---
14. Connect the supply. Adjust U ADJ in a clockwise direction until it is possible to check that the power supply is cut off within the following margins.	23,95 ± 0,05 V

DOCUMENT CHANGE HISTORY

Document change history

Version	Date	Change	Rationale
2.0	2005/05/09	*Document REGD-0000791_v2 replaces document 57019V0M000.*	Document reference changed.
3.0	May 2011	Added \`previous version 'data', 'change' and 'rationale' inside document change history section.	Update change history for previous document history modification.
3.0	May 2011	Document title.	Commercial reference substituted by project reference.
3.0	May 2011	*Document REGD-0000791_v3 replaces document REGD-0000791_v2.*	Document version changed.
3.0	May 2011	Header title.	Increase information to easy document search.
4.0	December 2014	New procedure	The adjustment/checking procedure for the current power supply unit is added to the one for the old model